Inverted MAML Card 1: Round 5 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1792, Mom=0.85, Clamp=±85 LSB

1. Round 5 Summary & Cumulative Comparison (R1–R5)

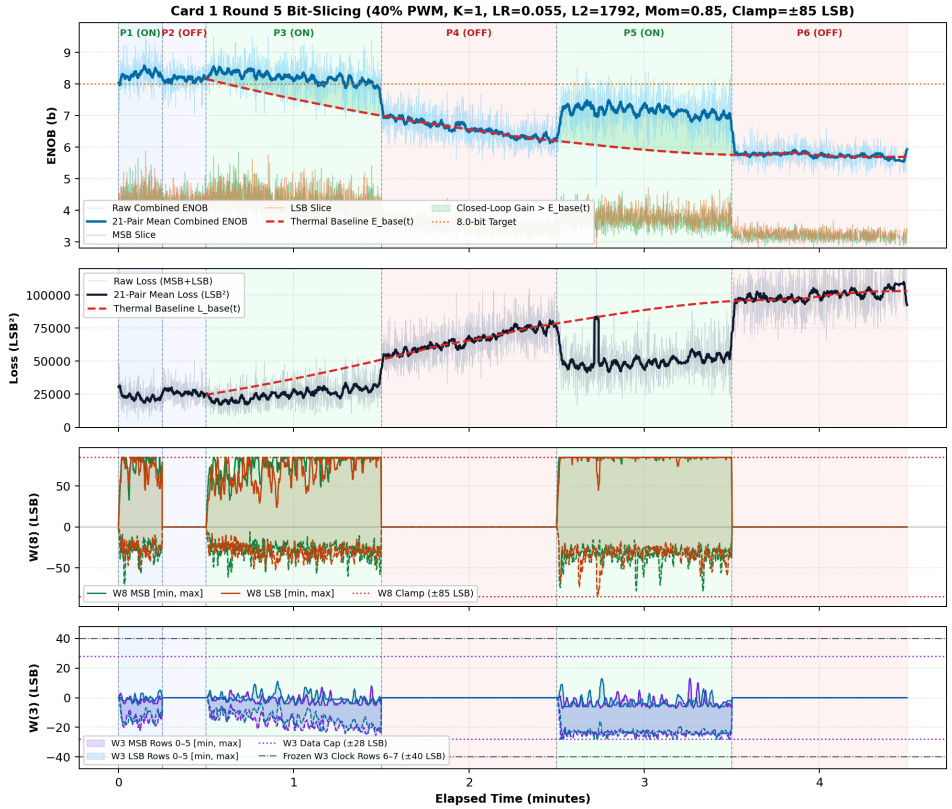
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 5 Detrended Trend Metrics:** Lift vs E_base = **+1.12 b**, Loss Ratio = **1.77x**, Climb = **+1.37 b**, Jitter = **0.42 b**, Recovery = **0.00**, Score = **+2.393**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1dc_fix	0.03	1024	0.9	±75	+1.00b	1.71x	+0.95b	0.40b	0.00	+1.799
R2dc_fix	0.035	1536	0.85	±75	+0.99b	1.69x	+1.16b	0.39b	0.00	+1.957
R3dc_fix	0.02	2560	0.8	±70	+0.04b	1.00x	+0.24b	0.34b	1.00	+0.204
R4dc_fix*	0.05	1792	0.85	±70	+1.08b	1.78x	+0.97b	0.40b	0.50	+2.480
R5dc_fix	0.055	1792	0.85	±85	+1.12b	1.77x	+1.37b	0.42b	0.00	+2.393

2. Phase-by-Phase Telemetry (4,077 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	320	7.98->8.34 (8.30b)	-	31391->21818	[85,85]	22	320
P2	Cold / Regler OFF	303	8.09->8.16 (8.15b)	-	28607->25302	[0,0]	0	0
P3	Warm / Regler ON	1043	8.22->8.06 (8.21b)	+0.87 b	24642->28681	[85,85]	26	1043
P4	Warm / Regler OFF	901	6.94->6.33 (6.59b)	+0.02 b	55223->74216	[0,0]	0	0
P5	Warm / Regler ON	814	6.52->7.04 (7.14b)	+1.24 b	68241->52505	[85,85]	28	814
P6	Warm / Regler OFF	696	5.82->5.70 (5.75b)	+0.01 b	94966->102515	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 5

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±85 LSB, while W(3) rows 0–5 stay within ±28 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 5 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

